IPS RISK AND RETURN OBJECTIVES

Learning Outcomes

- describe the major components of an IPS
- describe risk and return objectives and how they may be developed for a client
- explain the difference between the willingness and the ability (capacity) to take risk in analyzing an investor's financial risk tolerance

When constructing a portfolio for a client, it is important to ensure that the risk of the portfolio is suitable for the client. The IPS should state clearly the risk tolerance of the client. Risk objectives are specifications for portfolio risk that reflect the client's risk tolerance. Quantitative risk objectives can be absolute, relative, or a combination of the two.

Examples of an absolute risk objective would be a desire not to suffer any loss of capital or not to lose more than a given percentage of capital in any 12-month period. Note that these objectives are unrelated to investment market performance, good or bad, and are absolute in the sense of being self-standing. The fulfillment of such objectives could be achieved by not taking any risk—for example, by investing in an insured bank certificate of deposit at a creditworthy bank. If investments in risky assets are undertaken, however, such statements could be restated as a probability statement to be more operational (i.e., practically useful). For example, the desire not to lose more than 4% of capital in any 12-month period might be restated as an objective that with 95% probability the portfolio not lose more than 4% in any 12-month period. Measures of absolute risk include the variance or standard deviation of returns and **value at risk**.[2]

Some clients may choose to express relative risk objectives, which relate risk relative to one or more benchmarks perceived to represent appropriate risk standards. For example, investments in large-cap UK equities could be benchmarked to an equity market index, such as the FTSE 100 Index. The S&P 500 Index could be used as a benchmark for large-cap US equities; for investments with cash-like characteristics, the benchmark could be an interest rate such as Treasury bill rate. For risk relative to a benchmark, the measure could be **tracking risk**, or **tracking error**.[3] In practice, such risk objectives are used in situations where the total wealth management activities on behalf of a client are divided into partial mandates.

Other clients take both the investor's assets and liabilities into consideration when establishing an IPS risk objective. In some cases where the size, timing and/or relative certainty of future investor financial obligations are known, an IPS may be tailored to meet these objectives in what is called a **liability-driven investment (LDI)** approach. Examples of LDI include life insurance companies, defined benefit pension plans or an individual's budget after retirement. For example, a pension plan must meet the pension payments as they come due, and the risk objective will be to minimize the probability that it will fail to do so. A related return objective might be to outperform the discount rate used in finding the present value of liabilities over a multi-year time horizon.

When a policy portfolio (that is, a specified set of long-term asset class weightings and hedge ratios) is used, the risk objective may be expressed as a desire for the portfolio return to be within a band of plus or minus X% of the benchmark return calculated by assigning an index or benchmark to represent each asset class present in the policy portfolio. Again, this objective may be more usefully interpreted as a statement of probability—for example, a 95% probability that the portfolio return will be within X% of the benchmark return over a stated period. Example 1 reviews this material.

Rate Your Confidence
High
Medium
Low
Continue »

Category
Basics of Portfolio Planning and Construction

Related Questions:
Practice questions related to this topic

Types of Risk Objectives

A Japanese institutional investor has a portfolio valued at ¥10 billion. The investor expresses her first risk objective as a desire not to lose more than ¥1 billion in the coming 12-month period. She specifies a second risk objective of achieving returns within 4% of the return to the TOPIX stock market index, which is her benchmark. Based on this information, address the following:

1. **a.** Characterize the first risk objective as absolute or relative.
   b. Give an example of how the risk objective could be restated in a practical manner.

   Hide Solution

   Solution

   a. This is an absolute risk objective.
   b. This risk objective could be restated in a practical manner by specifying that the 12-month 95% value at risk of the portfolio must be no more than ¥1 billion.

2. **a.** Characterize the second risk objective as absolute or relative.
   b. Identify a measure for quantifying the risk objective.

   Hide Solution

   Solution

   a. This is a relative risk objective.
   b. This risk objective could be quantified using the tracking risk as a measure. For example, assuming returns follow a normal distribution, an expected tracking risk of 2% would imply a return within 4% of the index return approximately 95% of the time. Remember that tracking risk is stated as a one standard deviation measure.

A client's overall risk tolerance is a function of the client's ability to bear (accept) risk and her "risk attitude," which might be considered as the client's willingness to take risk. For ease of expression, from this point on we will refer to ability to bear risk and willingness to take risk as the two components of risk tolerance. Above-average ability to bear risk and above-average willingness to take risk imply above-average risk tolerance. Below-average ability to bear risk and below-average willingness to take risk imply below-average risk tolerance. These interactions are shown in Exhibit 1.

Exhibit 1:

Risk Tolerance

	Ability to Bear Risk	
Willingness to Take Risk	**Below Average**	**Above Average**
Below Average	Below-average risk tolerance	Resolution needed
Above Average	Resolution needed	Above-average risk tolerance

The *ability* to bear risk is measured mainly in terms of objective factors, such as time horizon, expected income, and level of wealth relative to liabilities. For example, an investor with a 20-year time horizon can be considered to have a greater ability to bear risk, other things being equal, than an investor with a 2-year horizon. This difference is

Similarly, an investor whose assets are comfortably in excess of their liabilities has more ability to bear risk than an investor whose wealth and expected future expenditure are more closely balanced. For example, a wealthy individual who can sustain a comfortable lifestyle after a very substantial investment loss has a relatively high ability to bear risk. A pension plan that has a large surplus of assets over liabilities has a relatively high ability to bear risk.

The *willingness* to take risk, or risk attitude, is a more subjective factor based on the client's psychology and perhaps also his current circumstances. Although the list of factors related to an individual's risk attitude remains open to debate, it is believed that some psychological factors, such as personality type, self-esteem, and inclination to independent thinking, are correlated with risk attitude. Some individuals are comfortable taking financial and investment risk, whereas others find it distressing. Although there is no single agreed-upon method for measuring risk tolerance, a willingness to take risk may be gauged by discussing risk with the client or by asking the client to complete a psychometric questionnaire. For example, financial planning academic John Grable and collaborators have developed 13-item and 5-item risk attitude questionnaires that have undergone some level of technical validation. The five-item questionnaire is shown in Exhibit 2.

Exhibit 2:

A Five-Item Risk Assessment Instrument

1. Investing is too difficult to understand.
   a. Strongly agree
   b. Tend to agree
   c. Tend to disagree
   d. Strongly disagree
2. I am more comfortable putting my money in a bank account than in the stock market.
   a. Strongly agree
   b. Tend to agree
   c. Tend to disagree
   d. Strongly disagree
3. When I think of the word "risk," the term "loss" comes to mind immediately.
   a. Strongly agree
   b. Tend to agree
   c. Tend to disagree
   d. Strongly disagree
4. Making money in stocks and bonds is based on luck.
   a. Strongly agree
   b. Tend to agree
   c. Tend to disagree
   d. Strongly disagree
5. In terms of investing, safety is more important than returns.

c. Tend to disagree

d. Strongly disagree

Source: Grable and Joo (2004).

The responses, a), b), c), and d), are coded 1, 2, 3, and 4, respectively, and summed. The lowest score is 5 and the highest score is 20, with higher scores indicating greater risk tolerance. For two random samples drawn from the faculty and staff of large US universities (n = 406), the mean score was 12.86 with a standard deviation of 3.01 and a median score (i.e., the middle score) of 13.

Note that a question, such as the first one in Exhibit 2, indicates that risk attitude may be associated with non-psychological factors (such as level of financial knowledge and understanding and decision-making style) as well as psychological factors.

The adviser needs to examine whether a client's ability to accept risk is consistent with the client's willingness to take risk. For example, a wealthy investor with a 20-year time horizon, who is thus able to take risk, may also be comfortable taking risk; in this case the factors are consistent. If the wealthy investor has a low willingness to take risk, there would be a conflict.

The conflict between ability and willingness to take risk can also arise in the institutional context. In addition, different stakeholders within the institution may take different views. For example, the trustees of a well-funded pension plan may desire a low-risk approach to safeguard the funding of the scheme and beneficiaries of the scheme may take a similar view. The sponsor, however, may wish a higher-risk/higher-return approach in an attempt to reduce future funding costs. When a trustee bears a fiduciary responsibility to pension beneficiaries and the interests of the pension sponsor and the pension beneficiaries conflict, the trustee should act in the best interests of the beneficiaries.

When both the ability and willingness to take risk are consistent, the investment adviser's task is the simplest. When ability to take risk is below average and willingness to take risk is above average, the investor's risk tolerance should be assessed as below average overall. When ability to take risk is above average but willingness is below average, the portfolio manager or adviser may seek to counsel the client and explain the conflict and its implications. For example, the adviser could outline the reasons why the client is considered to have a high ability to take risk and explain the likely consequences, in terms of reduced expected return, of not taking risk. The investment adviser, however, should not aim to change a client's willingness to take risk that is not a result of a miscalculation or misperception. Modification of elements of personality is not within the purview of the investment adviser's role. The prudent approach is to reach a conclusion about risk tolerance consistent with the lower of the two factors (ability and willingness) and to document the decisions made.

Example 2 is the first of a set that follows the analysis of an investment client through the preparation of the major elements of an IPS.

EXAMPLE 2

The Case of Henri Gascon: Risk Tolerance

1. Henri Gascon is an energy trader who works for a major French oil company based in Paris. He is 30 years old and married with one son, aged 5. Gascon has decided that it is time to review his financial situation and consults a financial adviser, who notes the following aspects of Gascon's situation:

   - Gascon's annual salary of €250,000 is more than sufficient to cover the family's outgoings.
   - Gascon owns his apartment outright and has €1,000,000 of savings.

equity markets will deliver positive returns over the long term.

- In the risk tolerance questionnaire, Gascon strongly disagrees with the statements that "making money in stocks and bonds is based on luck" and "in terms of investing, safety is more important than returns."
- Gascon expects that most of his savings will be used to fund his retirement, which he hopes to start at age 50.

Based only on the information given, which of the following statements is *most* accurate?

A. Gascon has a low ability to take risk but a high willingness to take risk.
B. Gascon has a high ability to take risk but a low willingness to take risk.
C. Gascon has a high ability to take risk and a high willingness to take risk.

Hide Solution

Solution:

C is correct. Gascon has a high income relative to outgoings, a high level of assets, a secure job, and a time horizon of 20 years. This information suggests a high *ability* to take risk. At the same time, Gascon is knowledgeable and confident about financial markets and responds to the questionnaire with answers that suggest risk tolerance. This result suggests he also has a high *willingness* to take risk.

EXAMPLE 3

The Case of Jacques Gascon: Risk Tolerance

1. Marie Gascon is so pleased with the services provided by her financial adviser that she suggests to her brother Jacques that he should also consult the adviser. Jacques thinks it is a good idea. Jacques, a self-employed computer consultant also based in Paris, is 40 years old and divorced with four children, aged between 12 and 16. The financial adviser notes the following aspects of Jacques' situation:

- Jacques' consultancy earnings average €40,000 per annum but are quite volatile.
- Jacques is required to pay €10,000 per year to his ex-wife and children.
- Jacques has a mortgage on his apartment of €100,000 and €10,000 of savings.
- Jacques has a good knowledge of financial matters and expects that equity markets will deliver very high returns over the long term.
- In the risk tolerance questionnaire, Jacques strongly disagrees with the statements "I am more comfortable putting my money in a bank account than in the stock market" and "When I think of the word 'risk', the term 'loss' comes to mind immediately."
- Jacques expects that most of his savings will be required to support his children at university.

Based only on the information given, which statement is correct?

A. Jacques has a low ability to take risk but a high willingness to take risk.
B. Jacques has a high ability to take risk but a low willingness to take risk.
C. Jacques has a high ability to take risk and a high willingness to take risk.

Hide Solution

Solution:

maintenance payments. His investment time horizon is approximately two to six years given the ages of his children and his desire to support them at university. This finely balanced financial situation and short time horizon suggests a low ability to take risk. In contrast, his expectations for financial market returns and risk tolerance questionnaire answers suggest a high willingness to take risk. The financial adviser may wish to explain to Jacques how finely balanced his financial situation is and suggest that, despite his desire to take more risk, a relatively cautious portfolio might be the most appropriate approach to take.

Return Objectives

A client's return objectives can be stated in a number of ways. Similar to risk objectives, return objectives may be stated on an absolute or a relative basis.

As an example of an absolute objective, the client may want to achieve a particular percentage rate of return. This objective could be a nominal rate of return or could be expressed in real (inflation-adjusted) terms.

Alternatively, the return objective can be stated on a relative basis—for example, relative to a benchmark return. The benchmark could be an equity market index, such as the S&P 500 or the FTSE 100, or a cash rate of interest such as the market reference rate (MRR). A relative return objective might be stated as, for example, a desire to outperform the benchmark index by one percentage point per year.

Some institutions also set their return objectives relative to a peer group or universe of managers—for example, an endowment aiming for a return that is in the top 50% of returns of similar institutions, or a private equity mandate aiming for returns in the top quartile among the private equity universe. This objective can be problematic when limited information is known about the investment strategies or the return calculation methodology being used by peers, and we must bear in mind the impossibility of *all* institutions being "above average." Furthermore, a good benchmark should be investable—that is, able to be replicated by the investor—and a peer benchmark typically does not meet that criterion.

In each case, the return requirement can be stated before or after fees. Care should be taken that the fee basis used is clear and understood by both the manager and client. The return can also be stated on either a pre- or post-tax basis when the investor is required to pay tax. For a taxable investor, the baseline is to state and analyze returns on an after-tax basis.

The return objective could be a required return—that is, the amount the investor needs to earn to meet a particular future goal—such as a certain level of retirement income.

The manager or adviser must ensure that the return objective is realistic. Care should be taken that client and manager are in agreement on whether the return objective is nominal (which is more convenient for measurement purposes) or real (i.e., inflation-adjusted, which usually relates better to the objective). It must be consistent with the client's risk objective (high expected returns are unlikely to be possible without high levels of risk) and also with the current economic and market environment. For example, 15% nominal returns might be possible when inflation is 10% but will be unlikely when inflation is 3%.

When a client has unrealistic return expectations, the manager or adviser will need to counsel her about what is achievable in the current market environment and within the client's tolerance for risk.

EXAMPLE 4

The Case of Marie Gascon: Return Objectives

which is 20 years from now. Her salary meets current and expected future expenditure requirements, but she does not expect to be able to make any additional pension contributions to her fund. Gascon sets aside €100,000 of her savings as an emergency fund to be held in cash. The remaining €900,000 is invested for her retirement.

Gascon estimates that a before-tax amount of €2,000,000 in today's money will be sufficient to fund her retirement income needs. The financial adviser expects inflation to average 2% per year over the next 20 years. Pension fund contributions and pension fund returns in France are exempt from tax, but pension fund distributions are taxable upon retirement.

1. Which of the following is closest to the amount of money Gascon will have to accumulate in nominal terms by her retirement date to meet her retirement income objective (i.e., expressed in money of the day in 20 years)?

   A. €900,000
   B. €2,000,000
   C. €3,000,000

   Hide Solution

   Solution:

   C is correct. At 2% annual inflation, €2,000,000 in today's money equates to €2,971,895 in 20 years measured in money of the day [€2,000,000 × $(1 + 2\%)^{20}$].

2. Which of the following is closest to the annual rate of return that Gascon must earn on her pension portfolio to meet her retirement income objective?

   A. 2.0%
   B. 6.2%
   C. 8.1%

   Hide Solution

   Solution:

   B is correct. €900,000 growing at 6.2% per year for 20 years will accumulate to €2,997,318, which is just above the required amount. (The solution of 6.2% comes from €2,997,318/€900,000 = $(1 + X)^{20}$, where X is the required rate of return.)

Discuss Filter

Discussion

What do you want to discuss... (required)

SC Hi all, what is the difference between the ability to bear risk and the willingness to take risk? Thank you,

Created 3 months ago by Selarath Chum 3 replies | Last Activity: 3 months ago

Show All replies Reply to this Comment

SC Hi all, in example 1, point 2.b. why risk is 2% and approximately 95% of the time? What does it mean? Thank you

Created 3 months ago by Selarath Chum 0 replies | Last Activity: 3 months ago

Reply to this Comment

Hi guys!

This implies that he is more focused on minimizing losses than maximizing returns, and hence, he has less Risk Tolerance. So it means he is less willing to take risk.

Could anyone explain it to me?

Created 8 months ago by Sayyed Ali Khurshid Khatib

3 replies | Last Activity: 5 months ago

Show All replies Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search